

2.1.6 Section Quiz

Question Review

An **internal threat** consists of someone like an employee that uses their authorized privileges to carry out an attack.

The term **hacker** is a general term used to describe any individual who uses their technical knowledge to gain unauthorized access to an organization.

A **hacktivist** is any individual whose attacks are politically motivated. Instead of seeking financial gain, hacktivists want to defame, shed light on, or cripple an organization or government. Hacktivists often work alone. Occasionally, they create unified groups with like-minded hackers. For example, the website wikileaks.org is a repository of leaked government secrets, some of which have been obtained by hacktivists.

insiders are one of the most dangerous and overlooked threats to an organization, you need to take the appropriate steps to protect against them, such as requiring mandatory vacations, creating and following onboarding and off-boarding procedure, employing the principle of least privilege, and having appropriate physical security controls in place.

script kiddies lack knowledge and sophistication, their attacks often seek to exploit well-known vulnerabilities in systems. As such, defense against script kiddies involves keeping systems up-to-date and using standard security practices.

the hacker is looking for an easy target and doesn't care what they are attacking. This is considered an **opportunistic attack**.

General attack strategies include the following steps:

- **Reconnaissance** is the process of gathering information about an organization, including system hardware information, network configuration, and individual user information.
- A **breach** is the penetration of system defenses. Breaches are achieved using the information gathered during reconnaissance.
- An **escalating privileges** attack is one of the primary objectives of an attacker, which can be achieved by configuring additional (escalated) rights to do more than breach the system.
- **Staging** is preparing a computer to perform additional tasks in the attack, such as installing software designed to attack other systems.
- An Exploit is used to take advantage of known vulnerabilities in software and systems. Types of **exploitation** include stealing information, denying services, crashing systems, and modifying information.

General defense methodologies include the following items:

- **Layering** is the process of implementing multiple security measures to protect the same asset. Defense in depth or security in depth is the premise that no single layer is completely effective in securing the assets. The most secure system/network has many layers of security and eliminates single points of failure.
- When using the **principle of least privilege**, users or groups are given only the access they need to do their job and nothing more. When assigning privileges, be aware that it is often easier to give a user more access when they need it than to take away privileges that have already been granted.
- Defensive layers should have **variety** and be diverse. Implementing multiple layers of the exact same defense does not provide adequate strength against attacks.
- **Randomness** relies on the constant change in personal habits and passwords to prevent anticipated events and exploitation.
- Security measures should provide protection, but not be so complex that you do not understand and use them.

Wanda being given access only to what she needs to do her job is an example of the principle of **least privilege**.

Reconnaissance is the phase of an attack where the attacker is gathering information about the target. This can be done electronically using scanning tools or even physically by going through dumpsters.